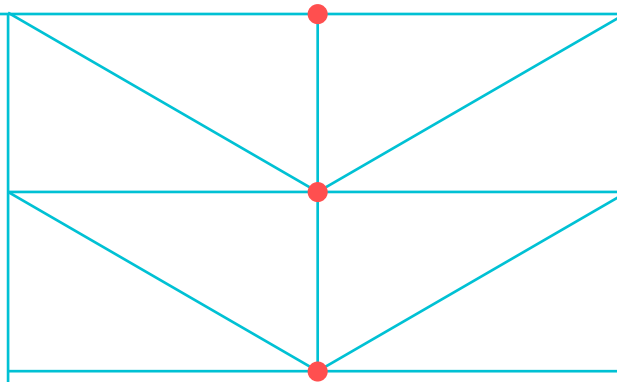


# ValidAgent

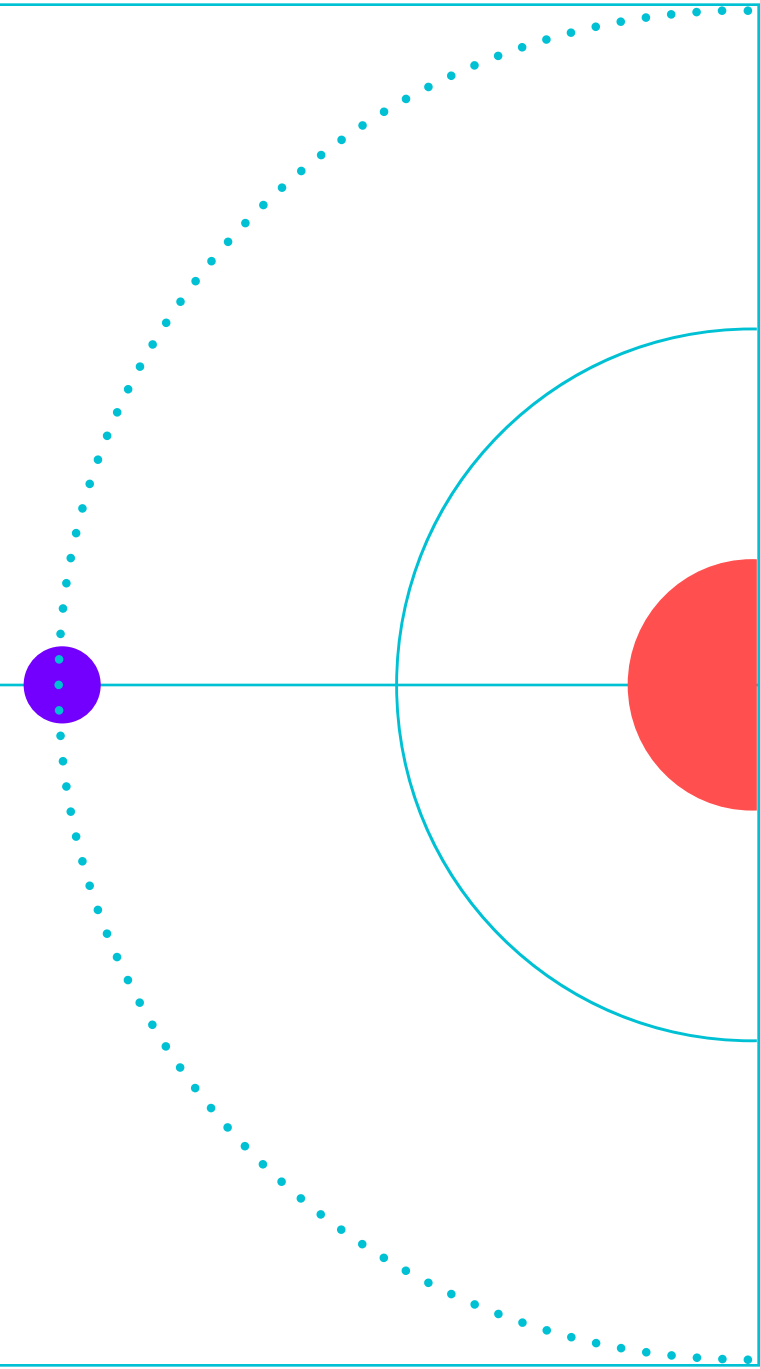
## Generative Agents

work in progress

**TUHH**  
Technische  
Universität  
Hamburg



**Leon Lührs, Christian Stindt von Dohm, Matthias Meyer**  
Institute of Management Accounting and Simulation



# Overview

## Overview

- Slide 1: Problem & Motivation: Toward Empirically Grounded Agent Identities
  - Slide 2: Repository Structure
  - Slide 3: Empirical Validation
  - Slide 4: Agent Selection & Export
  - Slide 5: Simulation Use Case
  - Slide 6: Vision & Community Input
- 
- <https://tuhh-maccs.github.io/validagent/>

# Toward Empirically Grounded Agent Identities

## Context

- Generative agents and LLM-based simulations are rapidly growing Many frameworks already support:
  - memory
  - planning
  - interaction
  - environments

## What is missing?

- reusable, validated agent identities
  - handcrafted personas framework
  - specific prompt engineering
  - low reproducibility
  - difficult reuse/comparison across studies
- ValidAgent aims to provide empirically grounded, reusable agent profiles for behavioral simulation research.

# Repository Structure (live on website)

## Two core entities

- **Agent Sets**

- collections of related agents
- shared validation context
- documented methodology

- **Agent Cards**

- individual agent identities
- traits, demographics, behaviors
- exportable simulation profiles

# Empirical Validation

**Problem:**

- Most persona repositories contain synthetic profiles.

**Our approach:**

- Agents are evaluated against observed human behavior.
- Die-roll honesty paradigm
- same task
- same conditions
- compare human vs agent behavior
- “These agents are not just descriptive — they are behaviorally benchmarked.”
- experiment modules
- demographics vs HEXACO vs freetext
- Correlations
- configurable inputs

# Agent Selection & Export

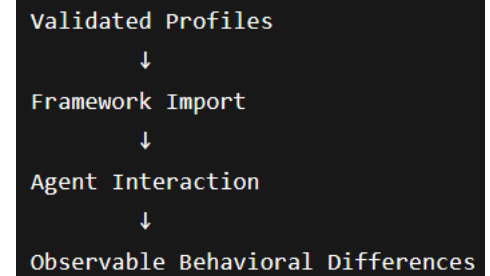
## Researchers can:

- inspect agents
  - select subpopulations
  - compare behavioral tendencies
  - export simulation-ready profiles
- 
- 2 honest
  - 2 dishonest
  - export JSON
- 
- Agents: 26, 42, 43, 51

| Scenario: Private Die-Roll Reporting Task [Run Round] |                                                                                                 |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| o Agent 26<br>"I report 1."                           | Prompt Inspector<br>_____<br>SYSTEM<br>You are an agent described<br>by the following traits... |
| o Agent 42<br>"I report 5."                           | PERSONA<br>High honesty-humility...                                                             |
| o Agent 43<br>"I report 4."                           | TASK<br>You rolled a 1...                                                                       |
| o Agent 51<br>"I report 5."                           |                                                                                                 |

# Simulation Use Case

- “The important point is not this toy simulation itself.”
- “The important point is that validated identities become reusable simulation primitives.”



# Vision & Community Input

## Near-Term Goals

- full empirical dataset release
- improved export compatibility
- framework integrations
- standardized metadata

## Long-Term Vision

- community-contributed agent sets
- reviewable validation pipelines
- reusable benchmark populations
- reproducible behavioral simulations
  
- Questions for the Community
- Which export formats/frameworks matter most?
- What metadata should validated agents contain?
- Would benchmark populations be useful for MAS research?



# Why Generative Agents?

## What are Generative Agents?

Generative agents are LLM-based computational personas that simulate human-like reasoning and decision-making in controlled environments.<sup>1</sup>

→ Their behavior changes depending on the persona information they are initialized with.

## Why use them?

### Experiments

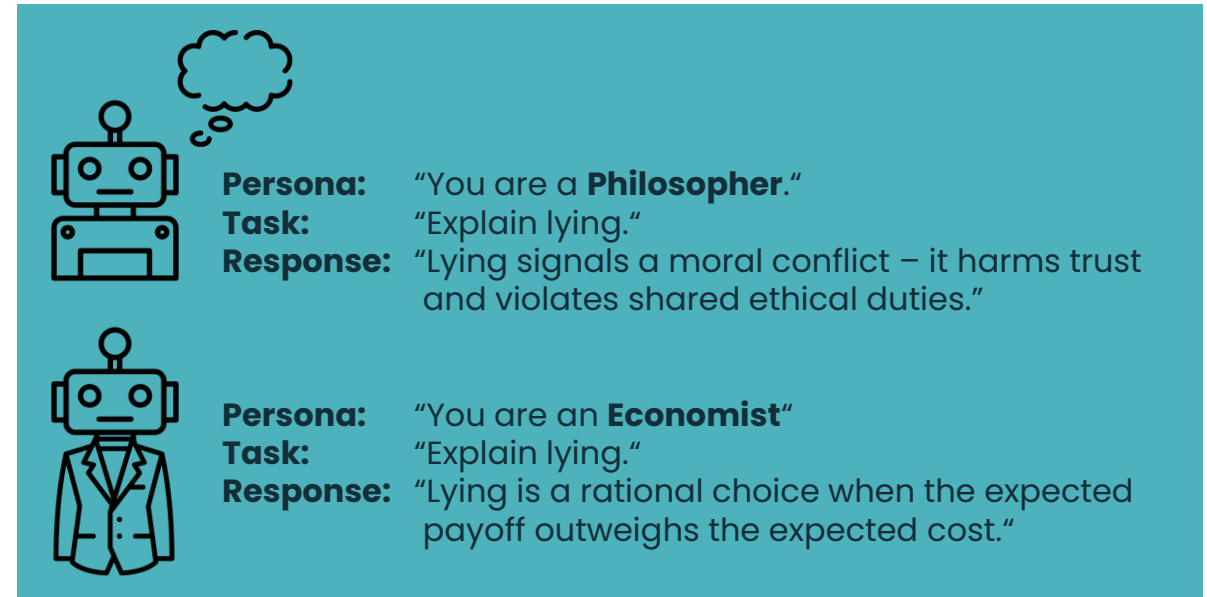
- Scalable & reproducible.
- Low cost & no ethical constraints.
- Targeted samples (e.g., specific traits or demographics).

### Simulation

- Rich behavioral agents.

## Why honesty?

- Central to management accounting and organizational behavior.
- Hard to observe directly in organizations.
- Experimental honesty tasks involve ethical constraints for humans.



The diagram illustrates two generative agent personas, each represented by a robot icon. The top robot is labeled 'Philosopher' and has a thought bubble above its head. The bottom robot is labeled 'Economist' and is wearing a suit. Both robots are shown with a 'Task' and a 'Response'.

| Persona            | Task             | Response                                                                              |
|--------------------|------------------|---------------------------------------------------------------------------------------|
| <b>Philosopher</b> | "Explain lying." | "Lying signals a moral conflict – it harms trust and violates shared ethical duties." |
| <b>Economist</b>   | "Explain lying." | "Lying is a rational choice when the expected payoff outweighs the expected cost."    |

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